

Stage II : Spatial Data Association of Camera and Radar Sensors

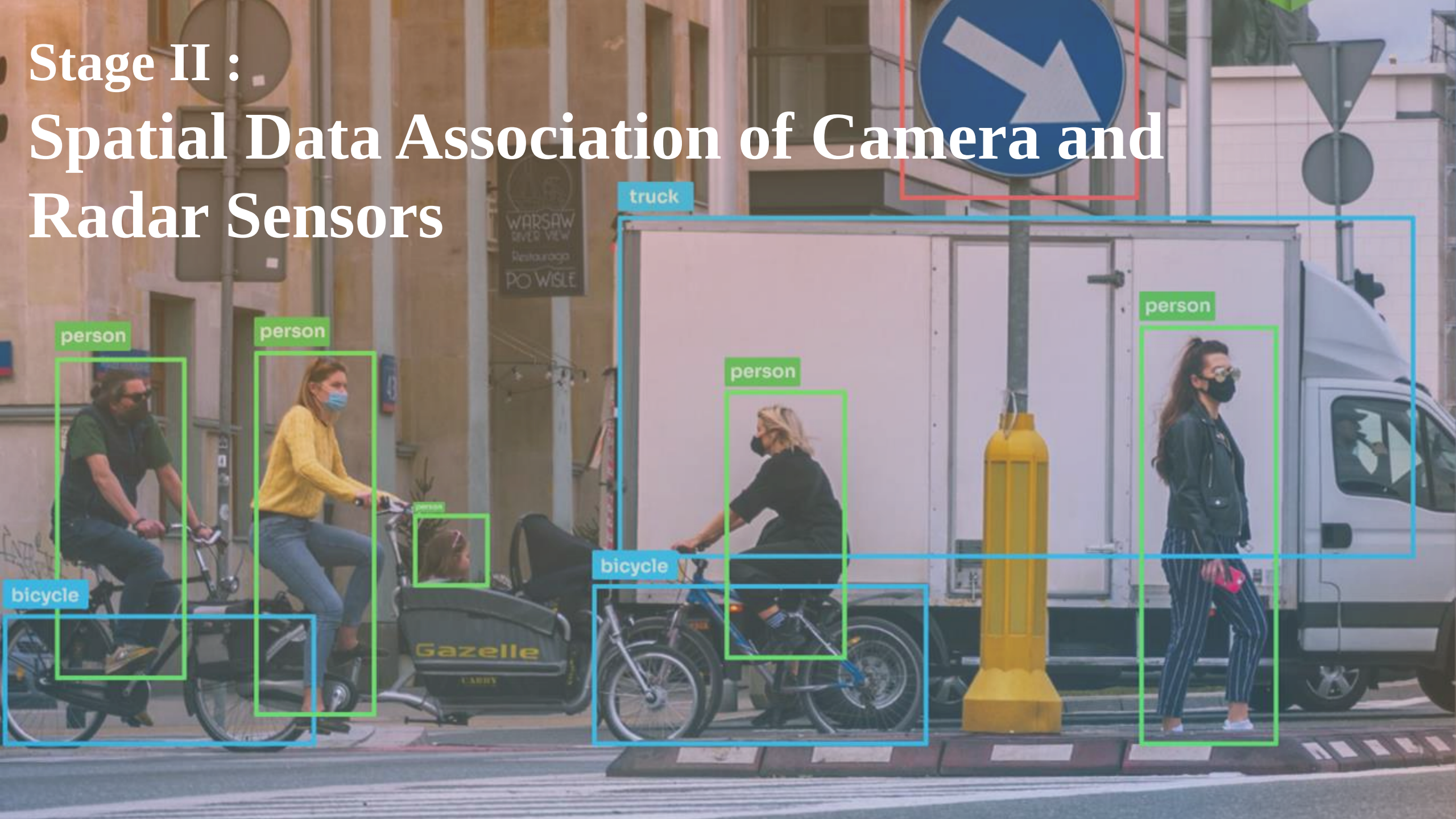
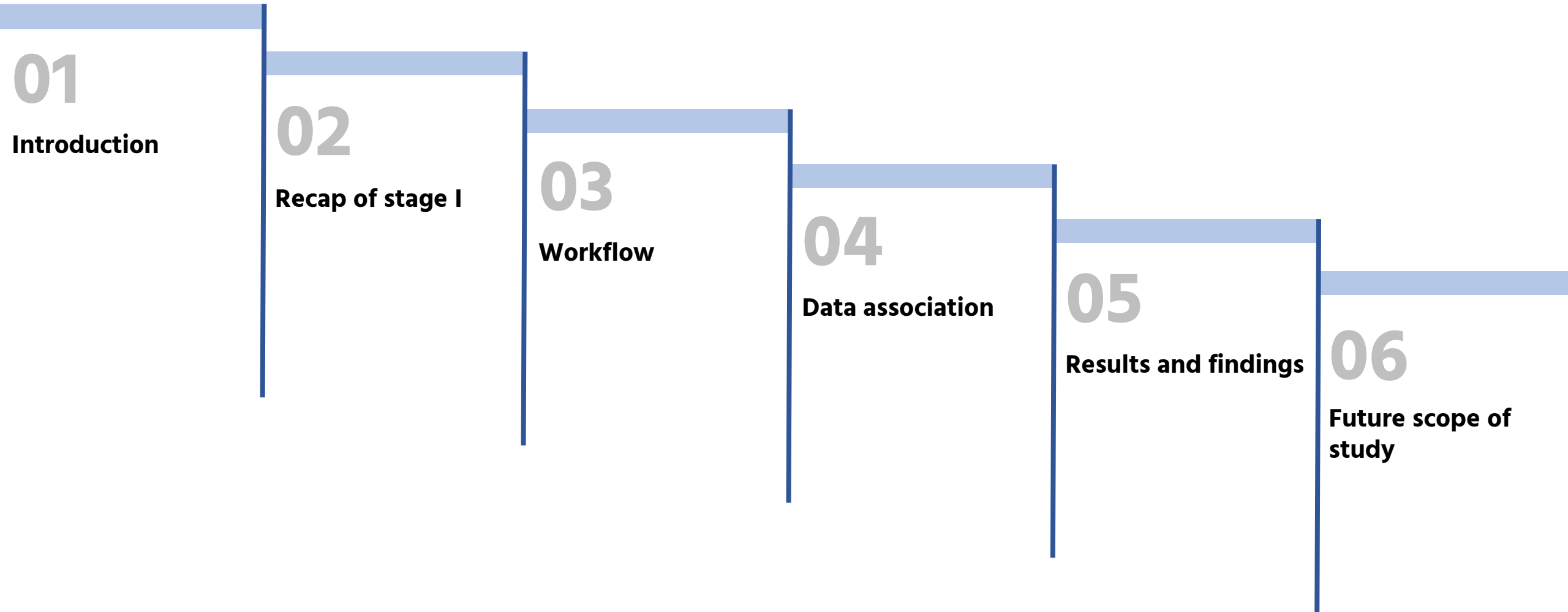


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Recap of stage I:

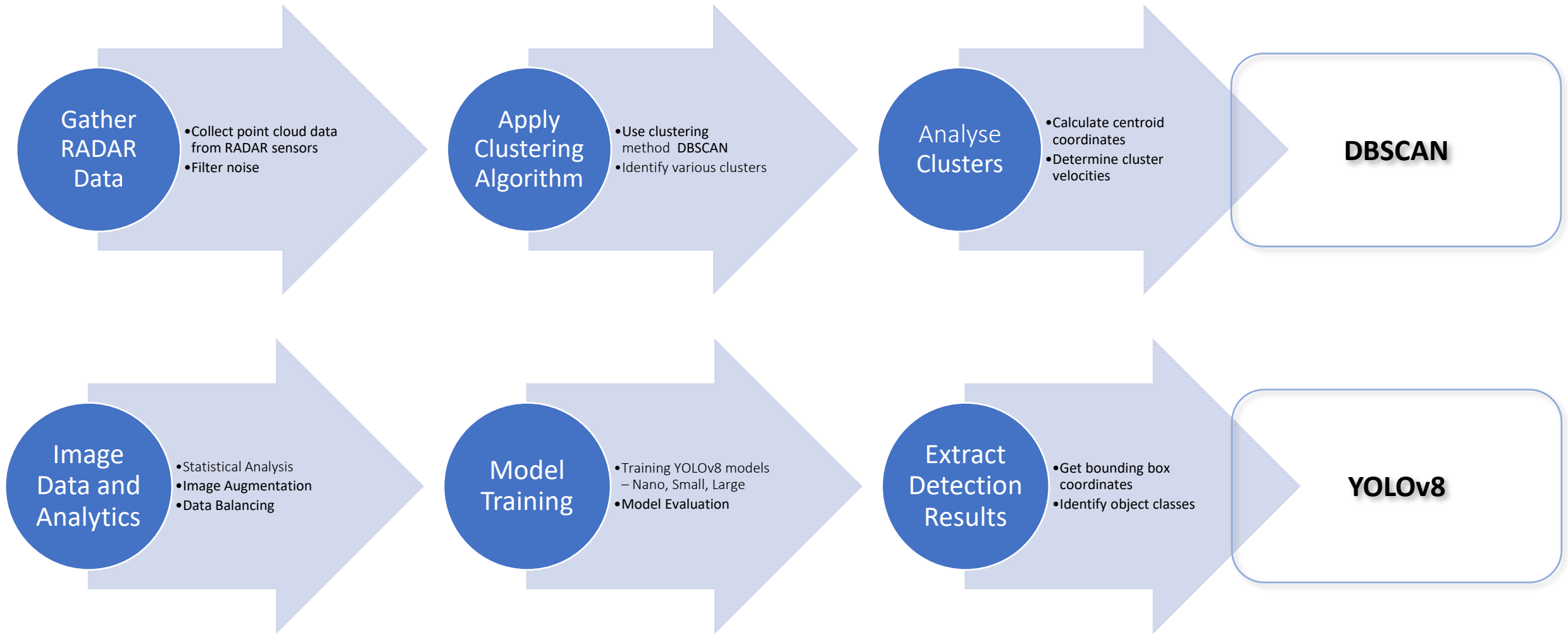


Fig 1. Brief recap of task accomplished in stage I

Improvements and additional changes:

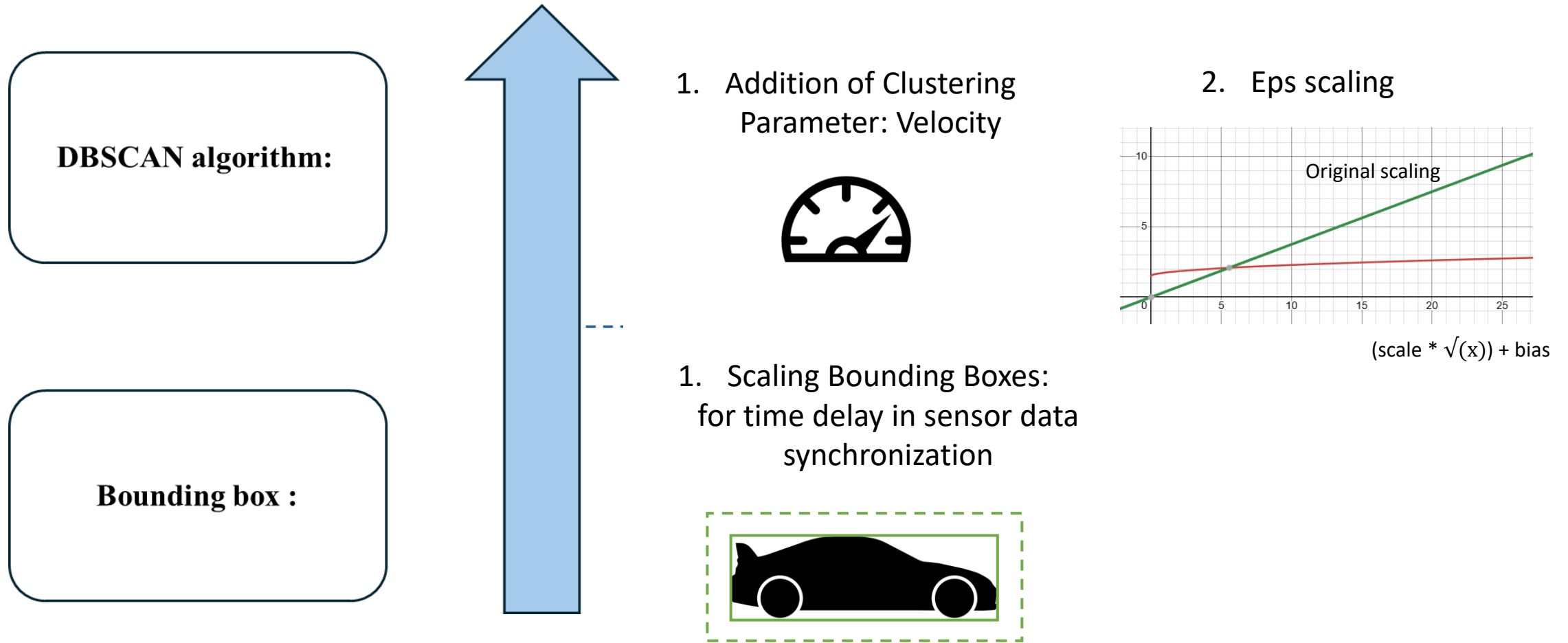


Fig 2. Changes and improvements made in previous work

Workflow pipeline :

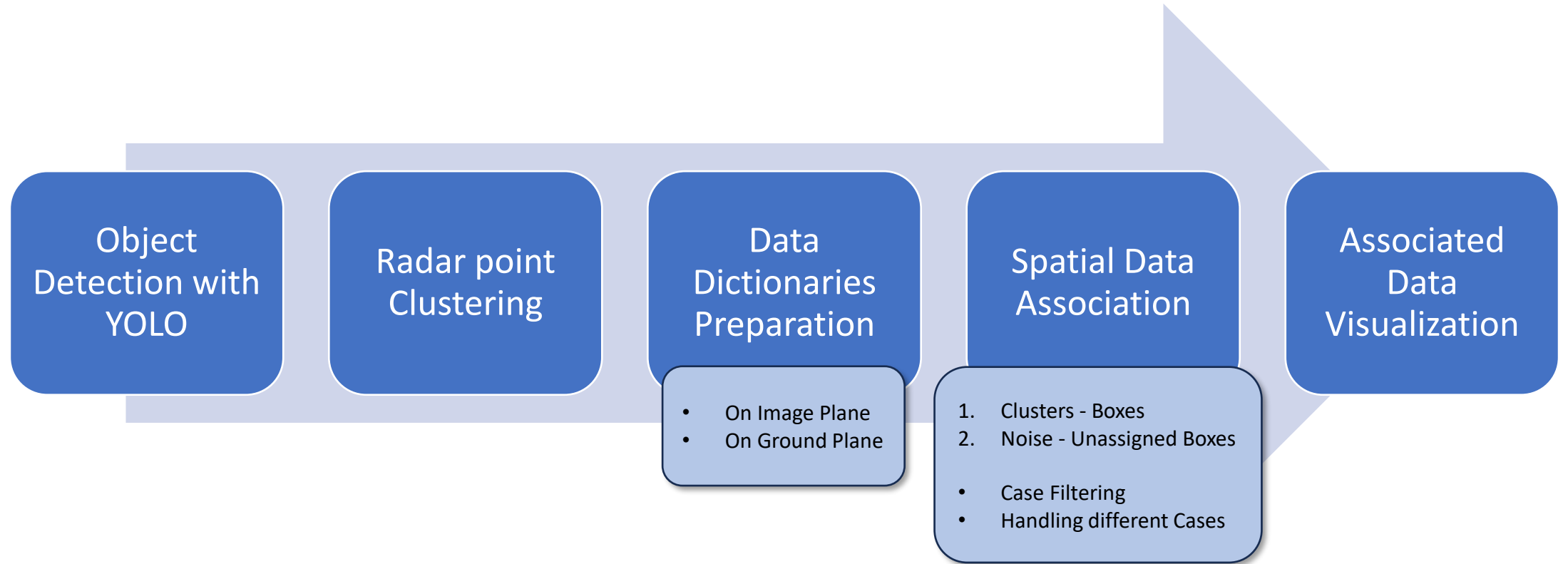


Fig 3. Workflow of stage II for spatial data association

Data Structure Example:

Radar Dictionaries:

```
{'clusters': [[[9.688504219055176, -1.177303433418274, -2.002965211868286],  
               [8.52904224395752, -1.177303433418274, -2.4082183837890625],  
               [9.115325927734375]]],  
 'noise': [[[18.304784774780273, 13.815544128417969, -0.3994672894477844],  
            [-0.12717212736606598]]]}
```

[[Centroid x y z]
 [minimum x y z]
 [doppler speed]]

[[Point x y z]
 [doppler speed]]

Data from YOLOv8:

```
[[ 3, 1163.5, 363.03, 1418.7, 550.87, 0.94738]]  
[[ class, x1, y1, x2, y2, confidence ]]
```

Data association:

Data
association and the
need for it:

Data association enables the correct matching of data points that refer to the same entity or event across different sources or time instances.

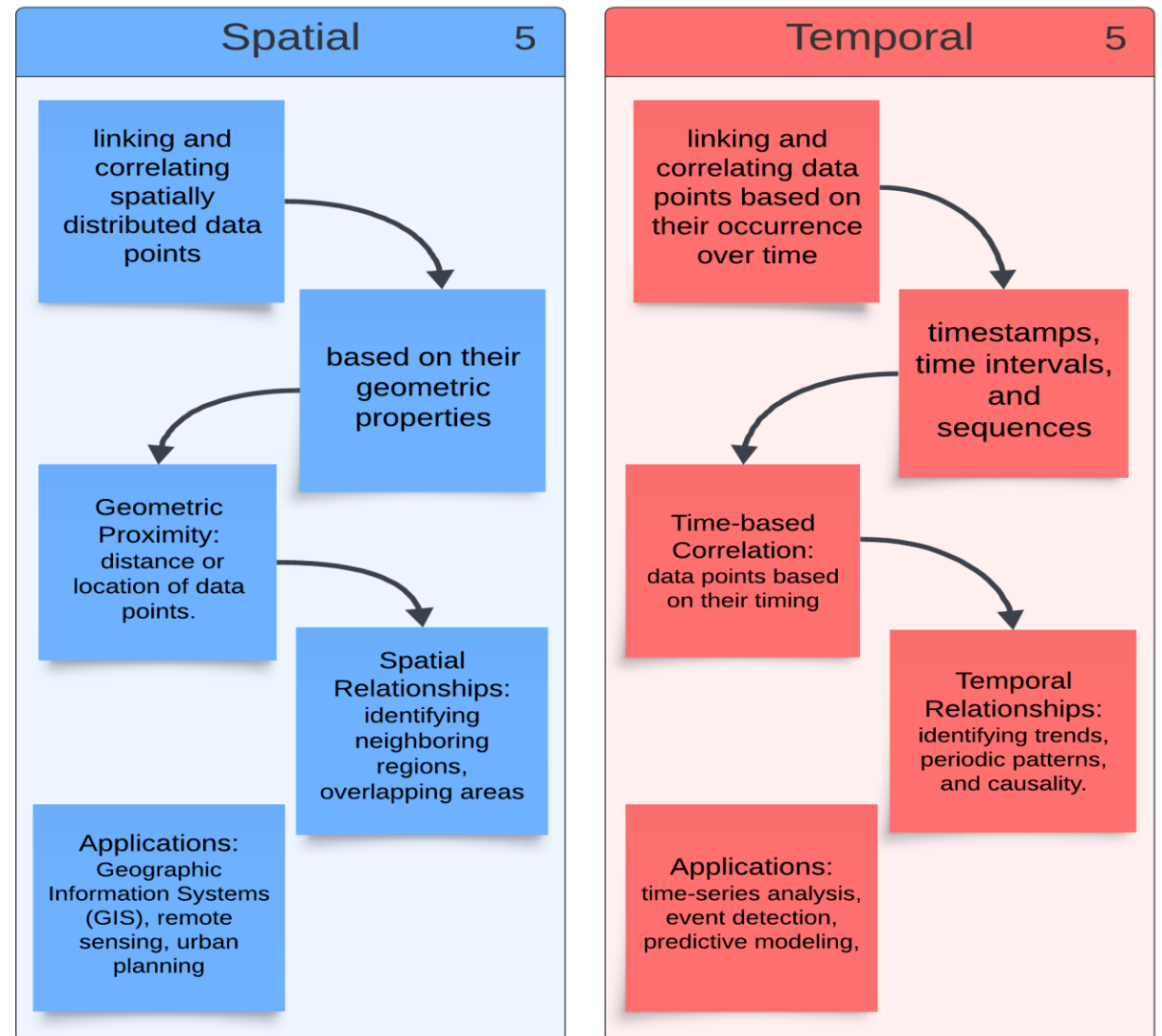


Fig 4. Spatial v/s Temporal Association

Data transformation of image and radar data :

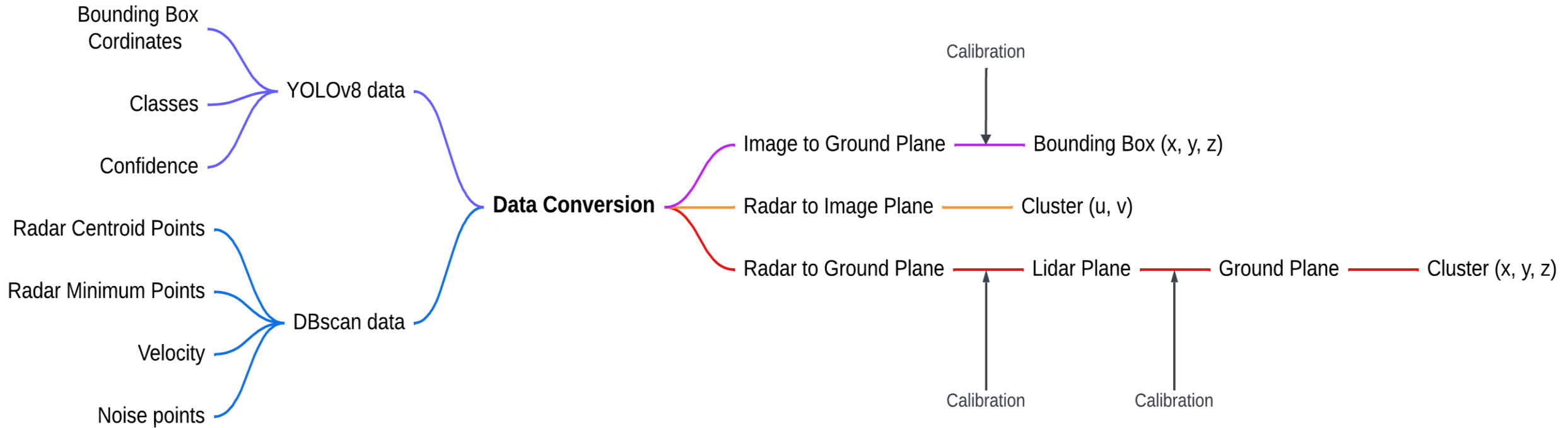
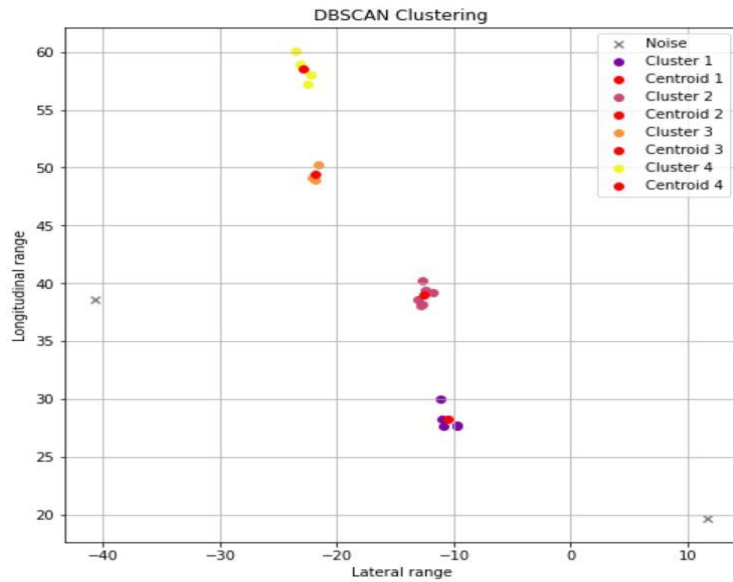
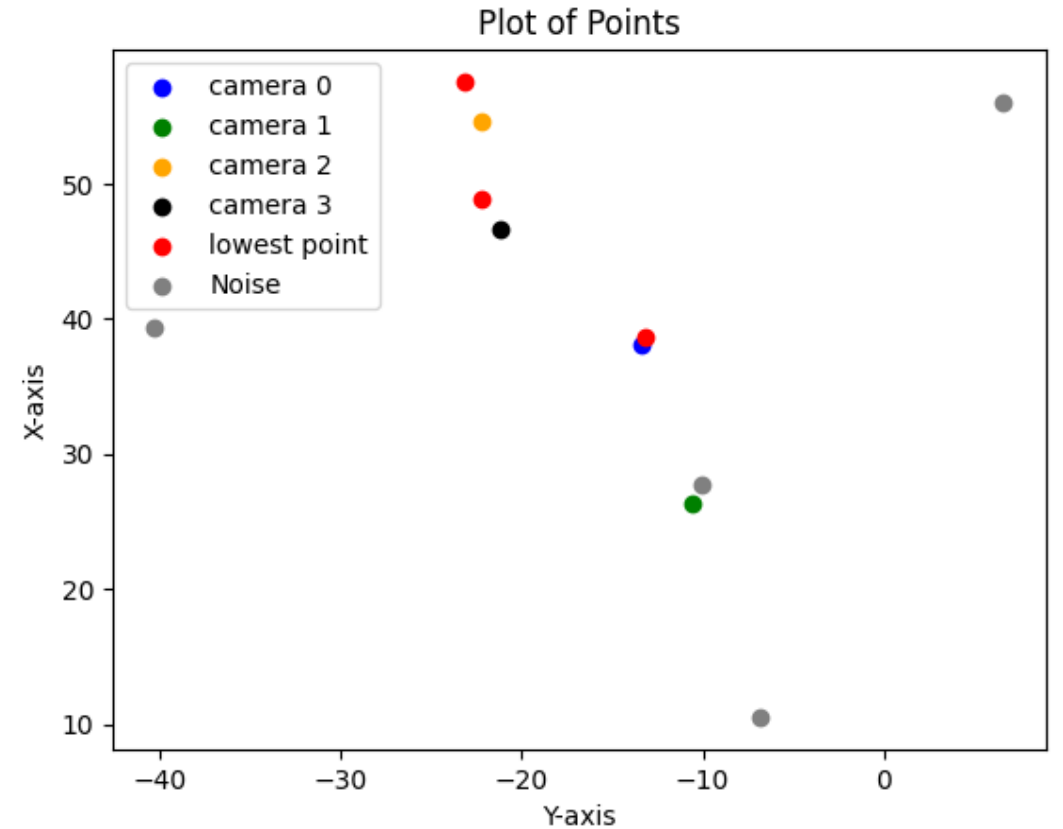


Fig 5. Data transformation workflow of radar and camera data

Radar cluster points from radar and image plane to ground plane :



Radar to lidar and lidar to ground

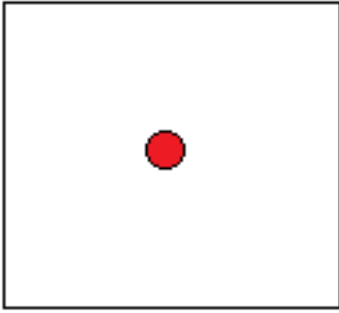


Camera to ground

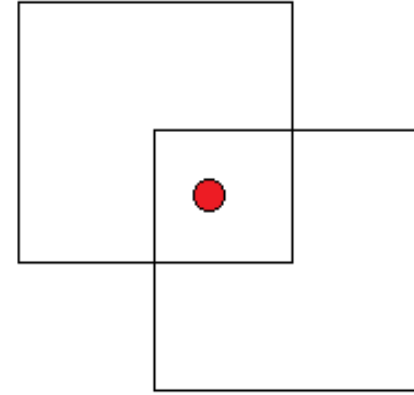


Fig 6. Conversion of image and radar planes to ground plane

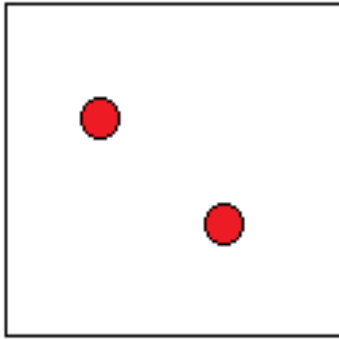
Filtering 4 different cases:



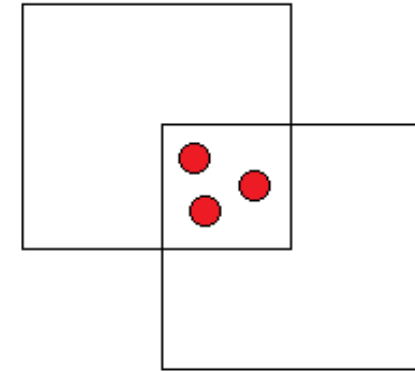
1) One Radar cluster to one bounding box



2) One radar cluster to many bounding box



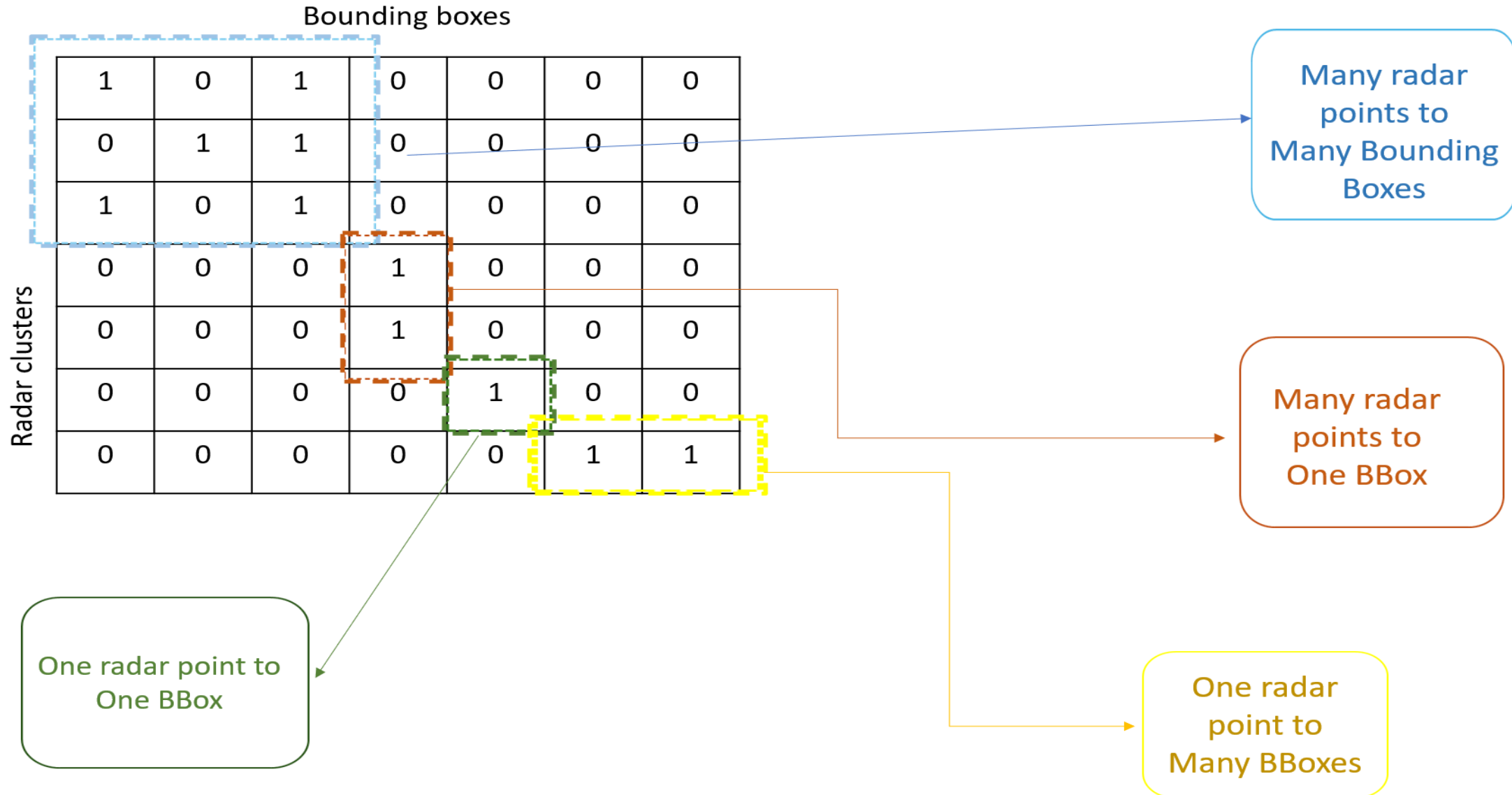
3) Many Radar clusters to one bounding box

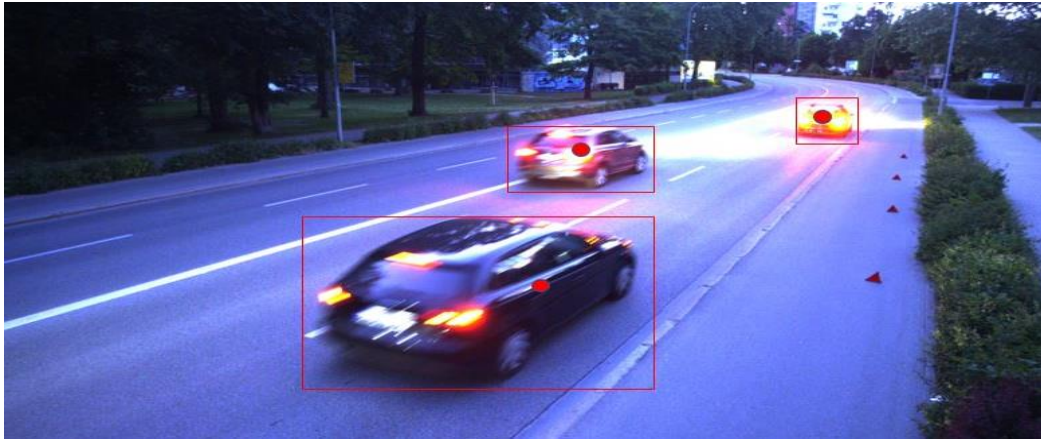


4) Many radar cluster to many bounding box

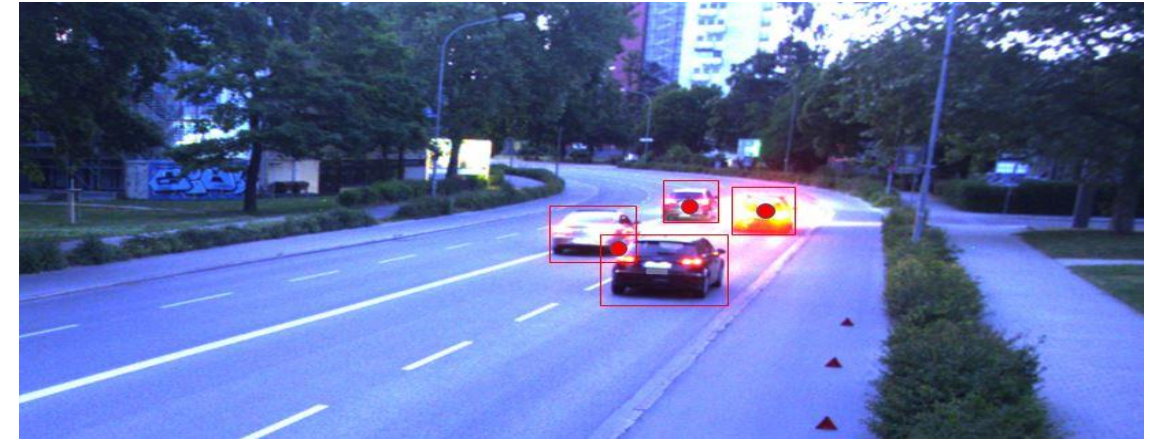
Fig 7. Different cases of scenarios

Filtering 4 different cases:

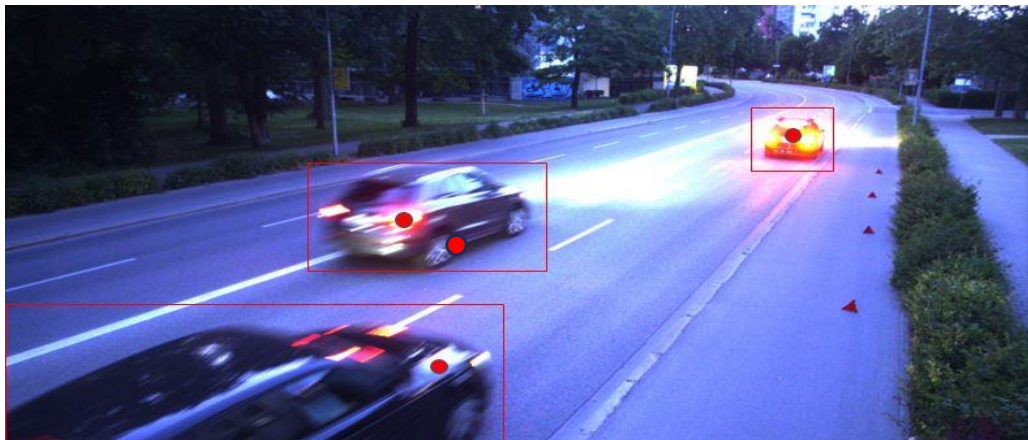




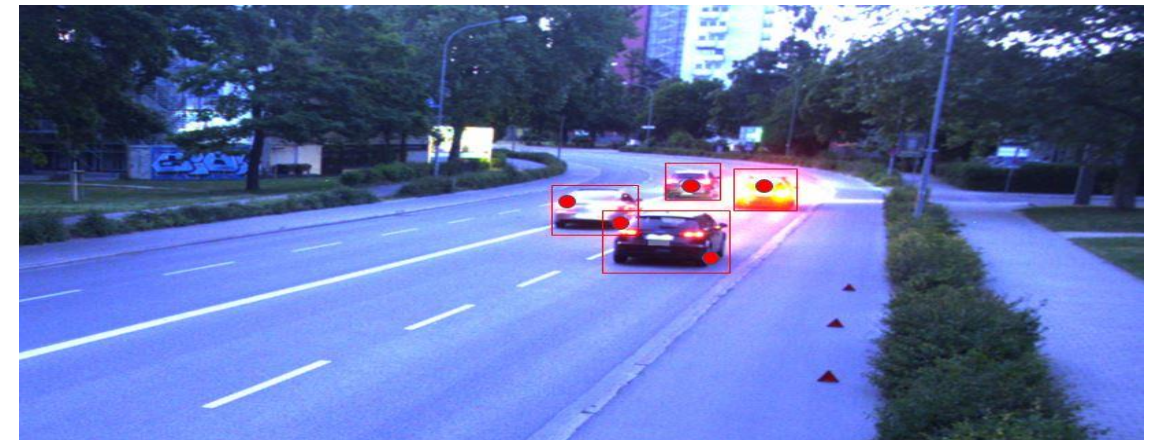
1) One Radar cluster to one bounding box



2) One radar cluster to many bounding box



3) Many Radar clusters to one bounding box



4) Many radar cluster to many bounding box

Fig 8. Different cases of scenarios from scene 15

Workflow of algorithm :

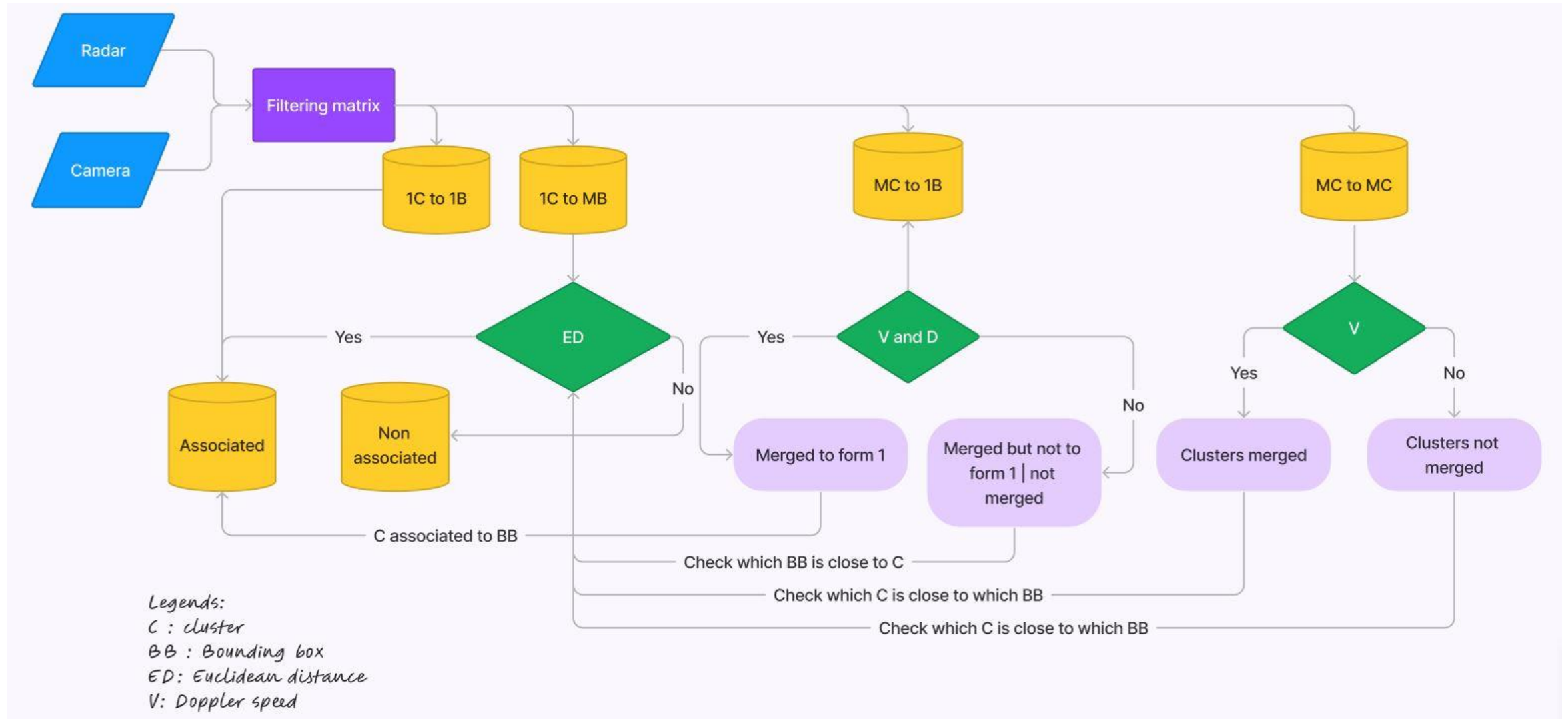
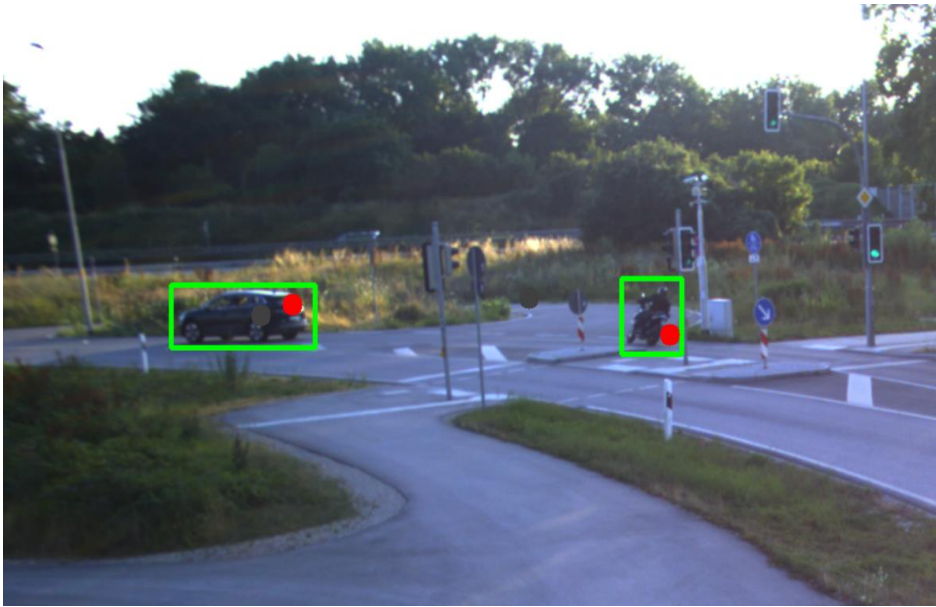


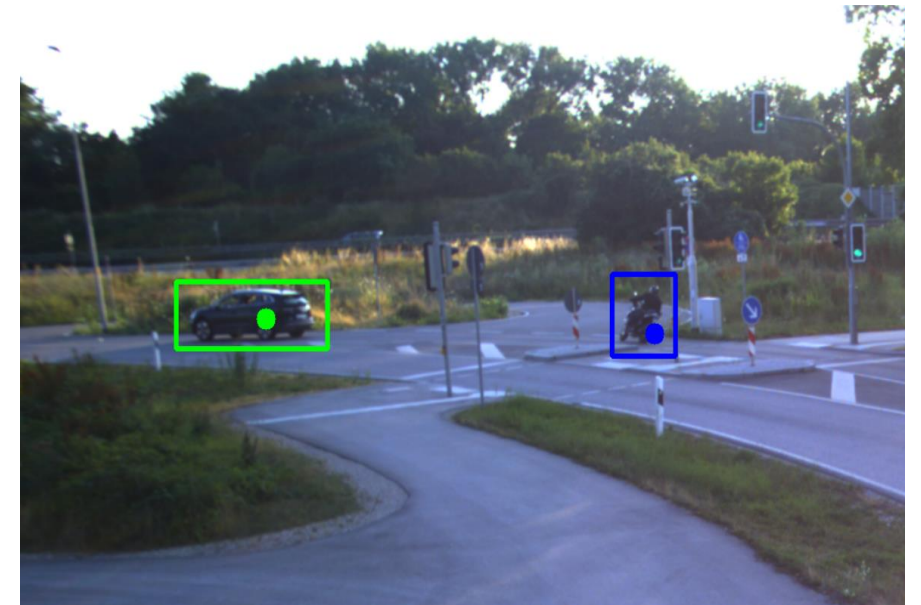
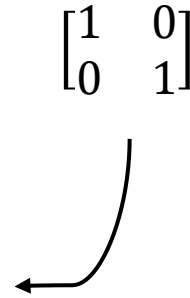
Fig . 9 Working structure of the algorithm

Handling different Cases: One radar cluster to one bounding box:

1. Comparing the coordinates of centroid with the coordinates of two corner points of the Bounding Box.
2. Centroid $\in (x, y)$ and bounding box $\in \{ (x_1, y_1), (x_2, y_2) \}$
3. For the condition to satisfy, $x_1 < x < x_2$ & $y_1 < y < y_2$



Before data association



After data association

Fig . 10 Case of 1 to 1 association from Scene 9

Handling different Cases: Many radar points to one bounding box :

1. We are comparing the velocity of all the cluster centroids.
2. All radar points whose velocities are in the threshold of ± 0.75 and within a certain value of distance from each other are merged.
3. In case of multiple merged clusters, we calculate the Euclidean distance of the points from lower centre point of Bounding box.
4. We associate the merged point with least Euclidean Distance to the Bounding Box.

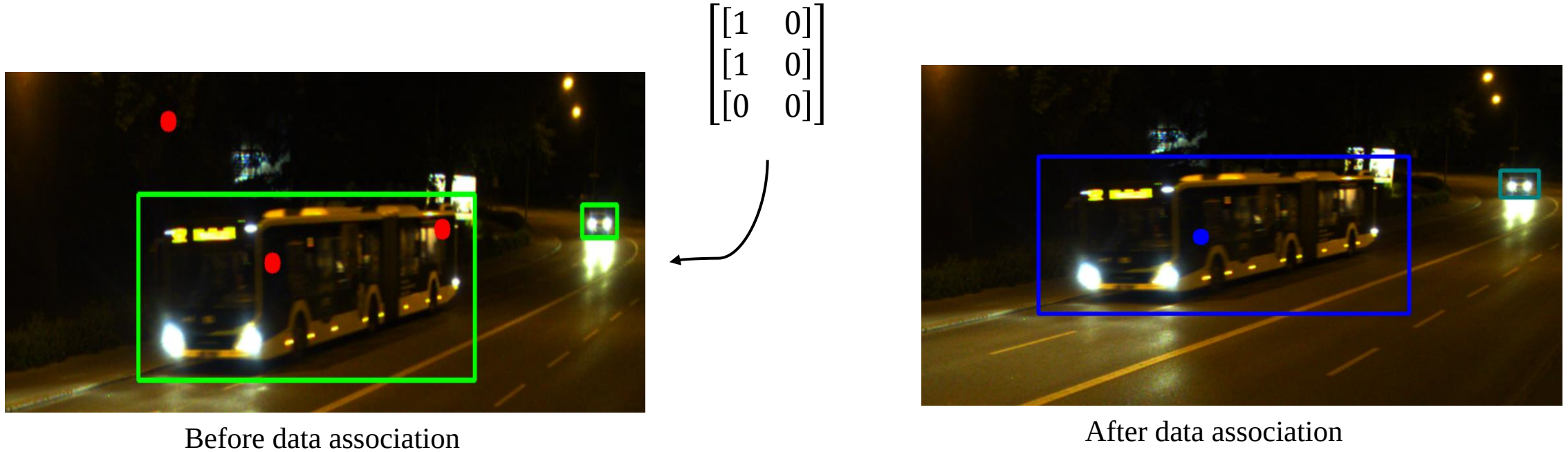
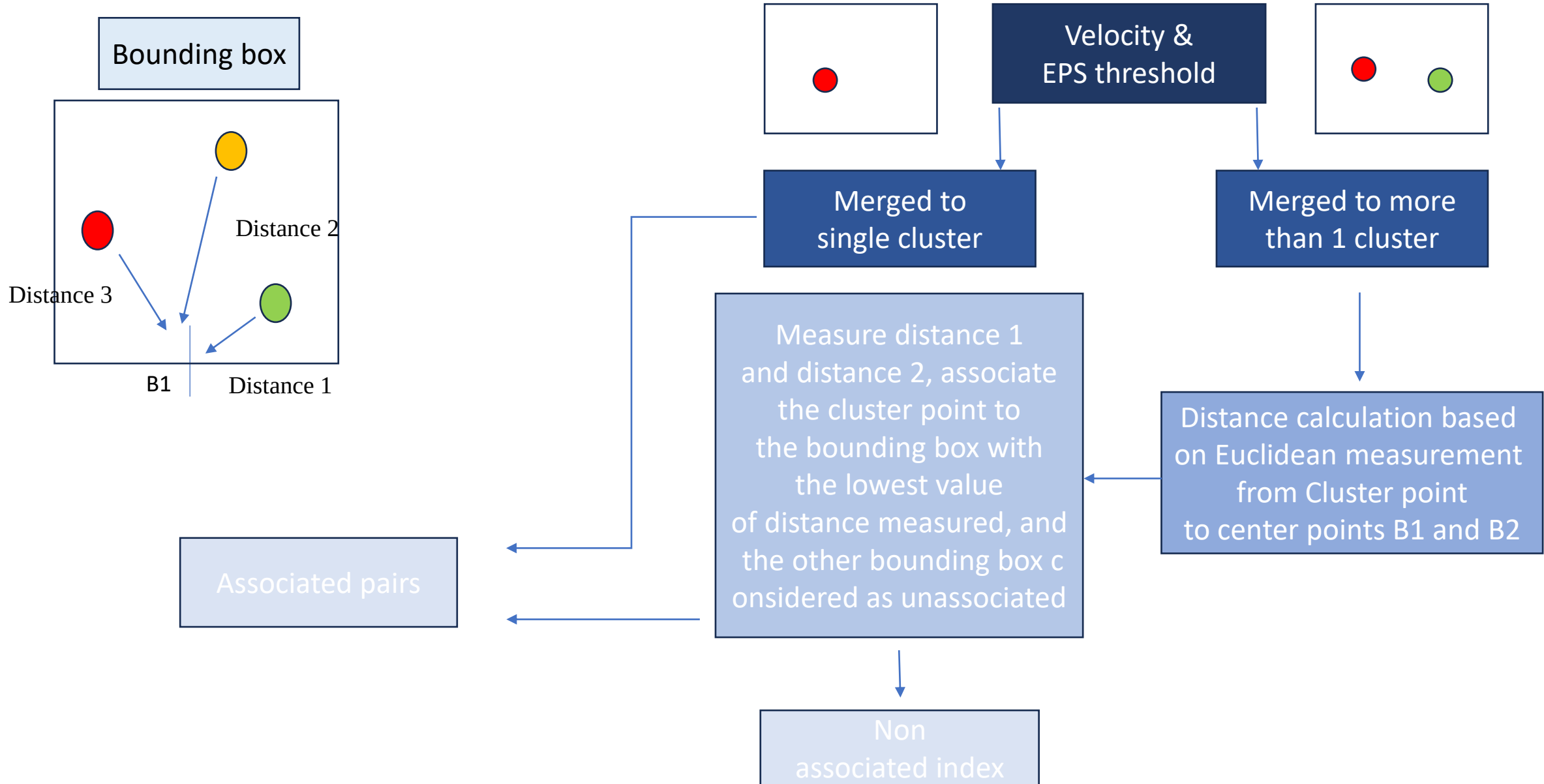


Fig . 11 Case of many radar to one bounding box from scene 22

Handling different Cases: Many clusters to one bounding box



Handling different Cases: One Radar point to Many Bounding Box

- 1) In this case we use the Euclidean distance and check the distance from the minimum cluster point with the bottom centre point of the bounding box.
- 2) The distance that has the least value is associated with that particular centroid.

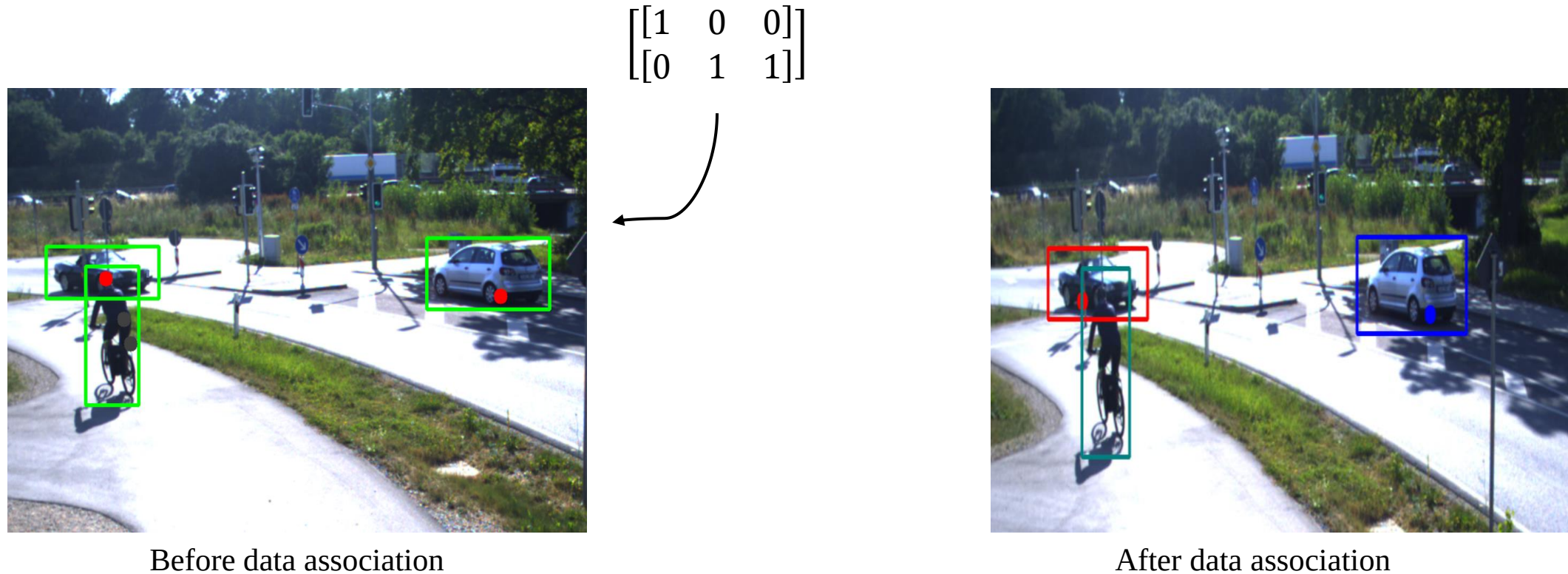
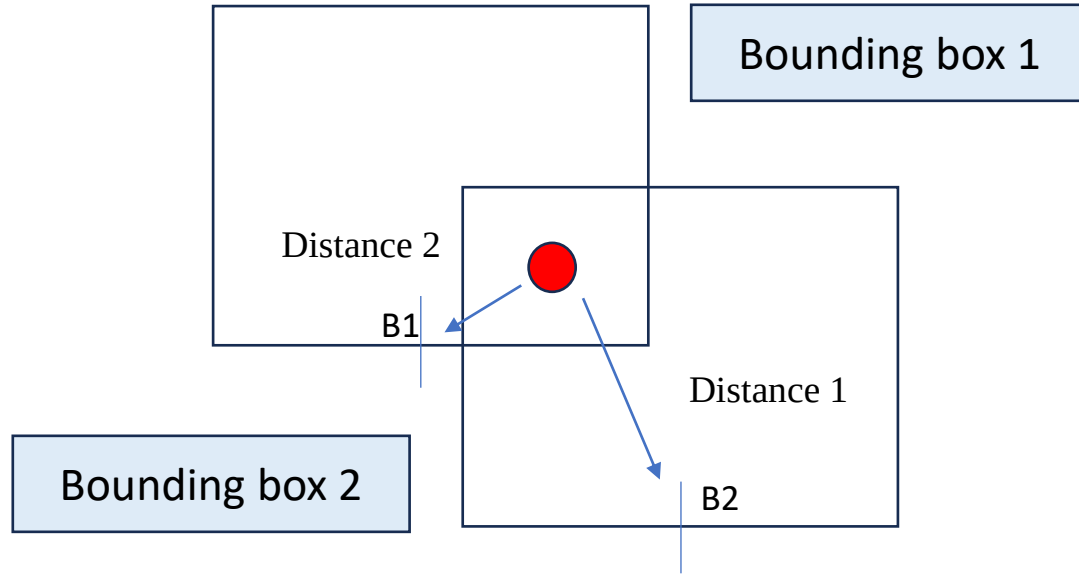


Fig . 12 Case of One Radar point to Many Bounding Box from scene 4

Handling different Cases: 1 radar cluster to many boxes



Distance calculation based on Euclidean measurement from Cluster point to center points B1 and B2

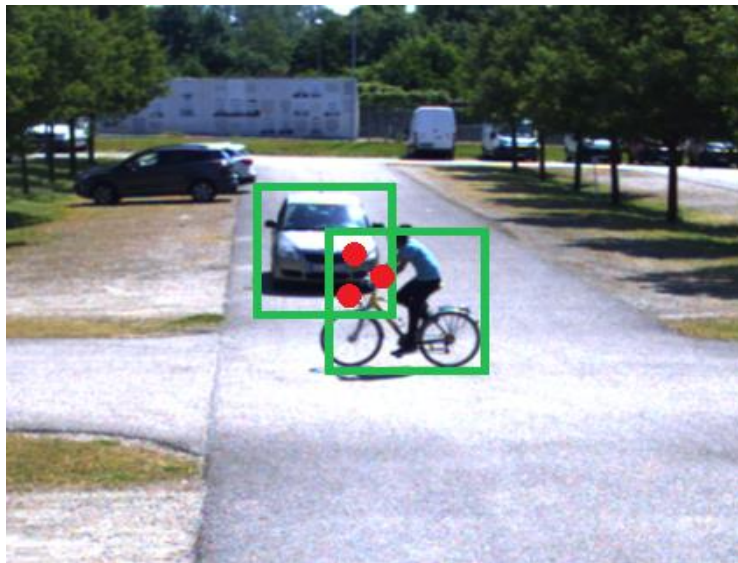
Measure distance 1 and distance 2, associate the cluster point to the bounding box with the lowest value of distance measured, and the other bounding box considered as unassociated

Associated pairs

Non associated index

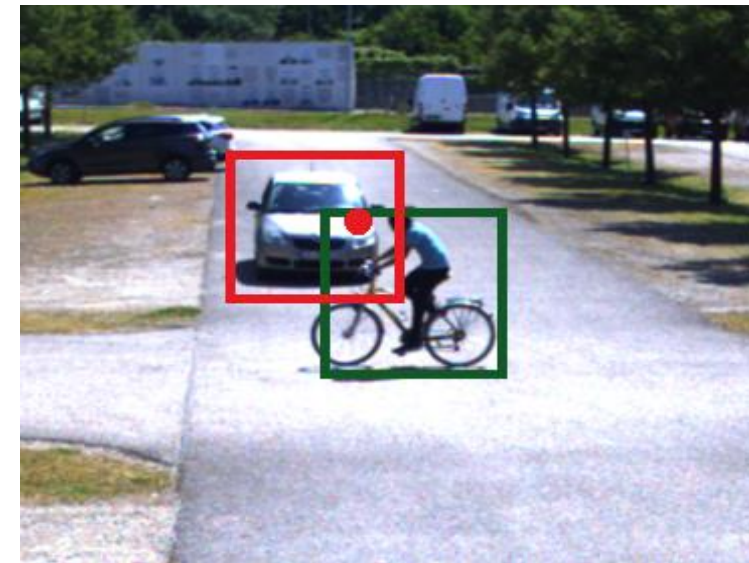
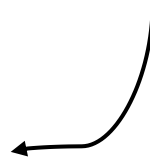
Handling different Cases: Many Radar points to Many Bounding Box

- 1) Measure distances from cluster points to bounding box centers using Euclidean distance, associating clusters to the nearest bounding box.
- 2) Clusters are merged based on velocity and EPS threshold, resulting in either equal or unequal numbers of clusters.
- 3) Identify associated pairs of clusters and bounding boxes, with unassociated elements noted separately.



Before data association

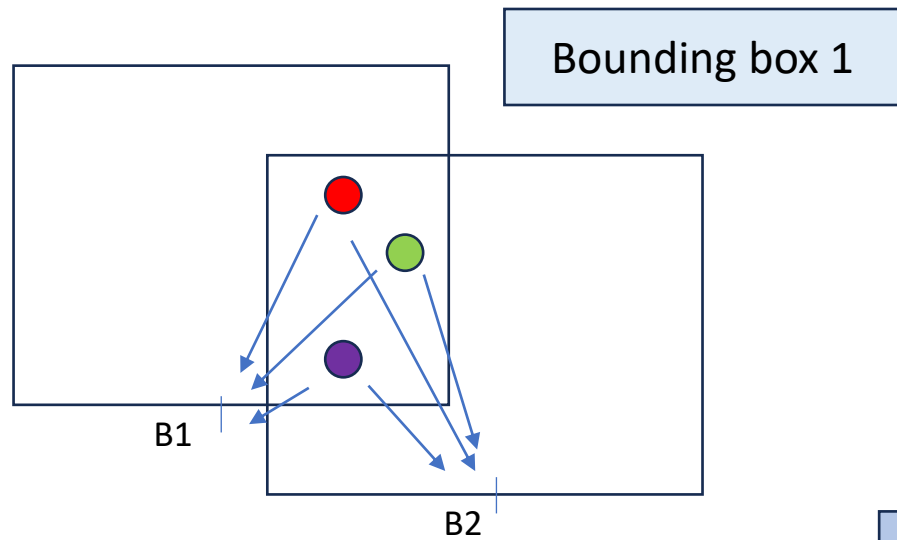
$$\begin{bmatrix} [1 & 1] \\ [1 & 1] \\ [1 & 1] \end{bmatrix}$$



After data association

Fig . 13 Case of One Radar point to Many Bounding Box from scene 21

Handling different Cases: Many clusters to many bounding box



Bounding box 2

Bounding box 1

Associated pairs

Non associated index

Measure distances from cluster to bounding box, associate the cluster point to the bounding box with the lowest value of distance measured, and the other cluster or bounding box considered as unassociated

Velocity & EPS threshold

Merged to form equal number of cluster

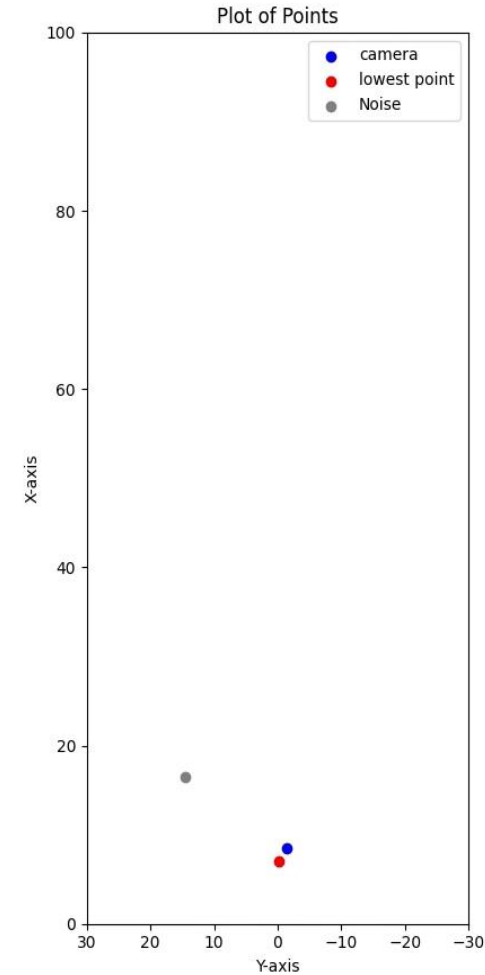
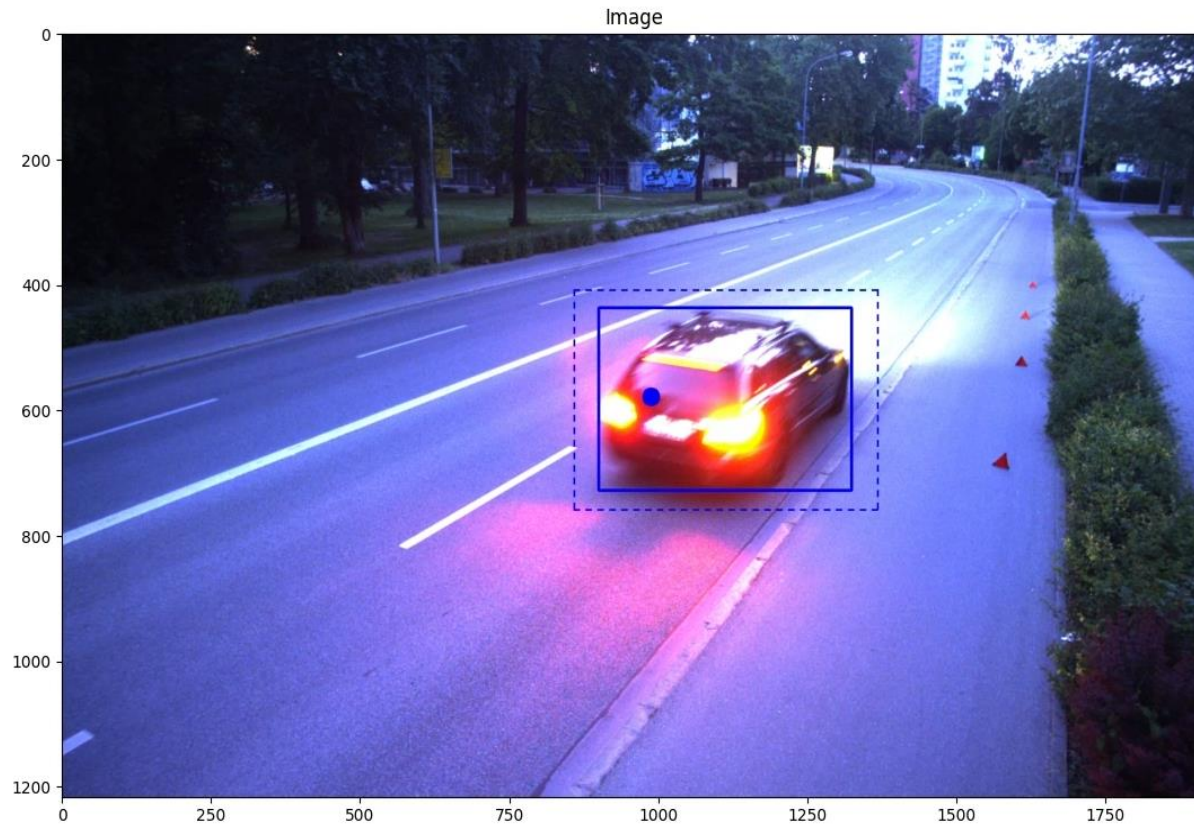
Merged to form unequal number of clusters

Distance calculation based on Euclidean measurement from Cluster point to center points B1 and B2

Output of spatial data association :

Returns a dictionary containing three keys : 'With Cluster' , ' With Noise' , 'Unassigned box'

These contains the pair of index from the matrix filtered in the very first step of data processing.



Future scope of study

- 1) Further study can be done on improving the clustering algorithm taking into consideration other parameters and information from camera sensor such as class to cluster the points.
- 2) Performing temporal data association with Extended Kalman filter for object tracking.
- 3) Using object class information from YOLO as input for cluster merging for different distance thresholds.

Learnings from the project :

- 1) Reading and extracting data from various formats.
- 2) Understanding how various sensors work.
- 3) Data analysis and pre-processing of data based on the requirement of the task.
- 4) Handling different cases and visually imagining problem statement.
- 5) Literature survey to understand similar cases of problems.
- 6) Visualizing data at each step to understand the results.
- 7) Creating a clean and readable code with scope of modularity.



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